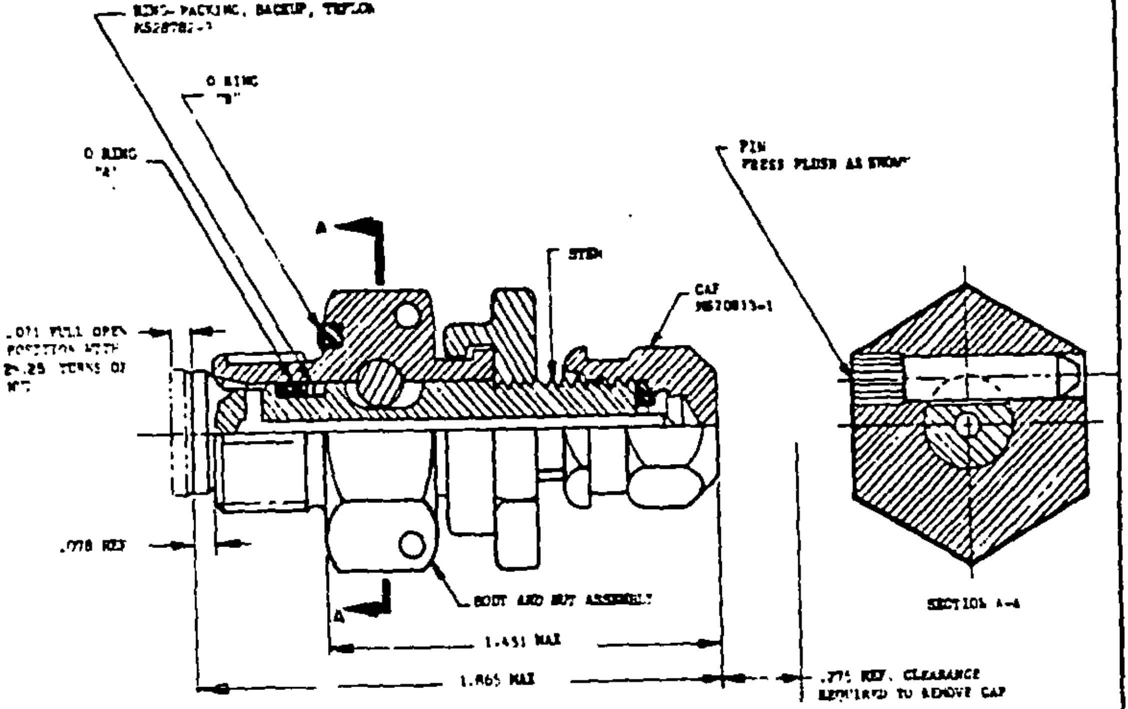
FED SUP CLASS
4820



MS PART NO	O-RING A	O-RING B
~~MS28889-1~~	~~AN6227-3~~	~~MS2...~~
MS28889-2	MS28775-008	MS28775-015

For New Design and Procurement use MS 28889-2

MS 28889-2 can replace MS 28889-1, but MS 28889-2 cannot be used in applications specifying MS 28889-2 except in instances where the environment for the application is below 160°F. Requisitions for MS 28889-1 should continue to be filled with MS 28889-1 until the stock is exhausted and then issue MS 28889-2. Requisitions for MS 28889-2 should be filled with only MS 28889-2.

MATERIAL: BODY, NUT AND STEM: CORROSION RESISTANT STEEL, AMS 5640 and AMS 5641, CLASS 303 or 303SE.
PIN RETAINER: CORROSION RESISTANT STEEL AMS 5613 and AMS 5610, CLASS 410 or 416.

HEAT TREAT: PIN RETAINER; 160,000 TENSILE STRENGTH (APPROXIMATELY), ROCKWELL C-36 TO C-40.

PASSIVATE: PIN RETAINER: SOLUTION, 18 TO 22 PERCENT NITRIC ACID (WT), 1.80 TO 2.50 PERCENT SODIUM DICHROMATE AT 120° TO 130°F FOR 15 TO 30 MINUTES, WASH IN WATER.

REMOVE ALL BURRS AND SHARP EDGES.

FINISH: BODY, NUT, PIN RETAINER; ALL MACHINED SURFACES 125 RHR MAX, STANDARD ANSI B46.1, UNLESS OTHERWISE SPECIFIED.
STEM: ALL SURFACES 125 RHR MAX, STANDARD ANSI B46.1, UNLESS OTHERWISE SPECIFIED.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ± .005, ANGLES ± .50°.

MAXIMUM WORKING PRESSURE: 5,000 PSI.

ASSEMBLY SHALL BE FURNISHED WITH CAP, MS20813-1.

FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.

CERTAIN PROVISIONS OF THIS STANDARD ARE THE SUBJECT OF INTERNATIONAL STANDARDIZATION AGREEMENTS ASCC AIR STD 25/35 & 25/36. WHEN REVISION OR CANCELLATION OF THIS STANDARD IS PROPOSED, THE DEPARTMENTAL CUSTODIANS WILL INFORM THEIR RESPECTIVE DEPARTMENTAL STANDARDIZATION OFFICE SO THAT APPROPRIATE ACTION MAY BE TAKEN RESPECTING THE INTERNATIONAL AGREEMENT CONCERNED.

RECLAIMED MATERIALS. THE USE OF RECLAIMED MATERIALS SHALL BE ENCOURAGED TO THE MAXIMUM EXTENT POSSIBLE.

APPROVED 6 JUN 55
REVISED Ⓙ 27 MAY 77 Ⓚ 24 JUNE 84 Ⓛ 27 NOV 87

P. A. USAF-71 Other Cust ARMY-AV NAVY-AS USAF-99	INTERNATIONAL INTEREST ASCC STDS 25/35 25/36	TITLE VALVE, AIR, HIGH PRESSURE CHARGING, 5000 PSI	MILITARY STANDARD MS 28889
PROCUREMENT SPECIFICATION MIL-V-6164	SUPERSEDES USAF DWGS 54B432 THROUGH 54B437		SHEET 1 OF 3

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

AMSC N/A

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

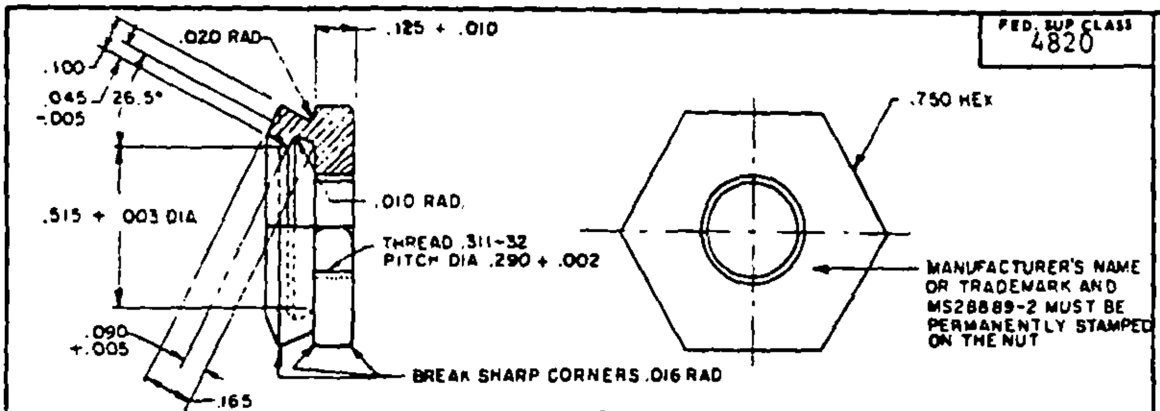
REVIEWER ACTIVITIES:
Army-AL, MI, ME
Navy-AS
Air Force-11

USER ACTIVITIES:
Army-AT
Defense Construction Supply-CS

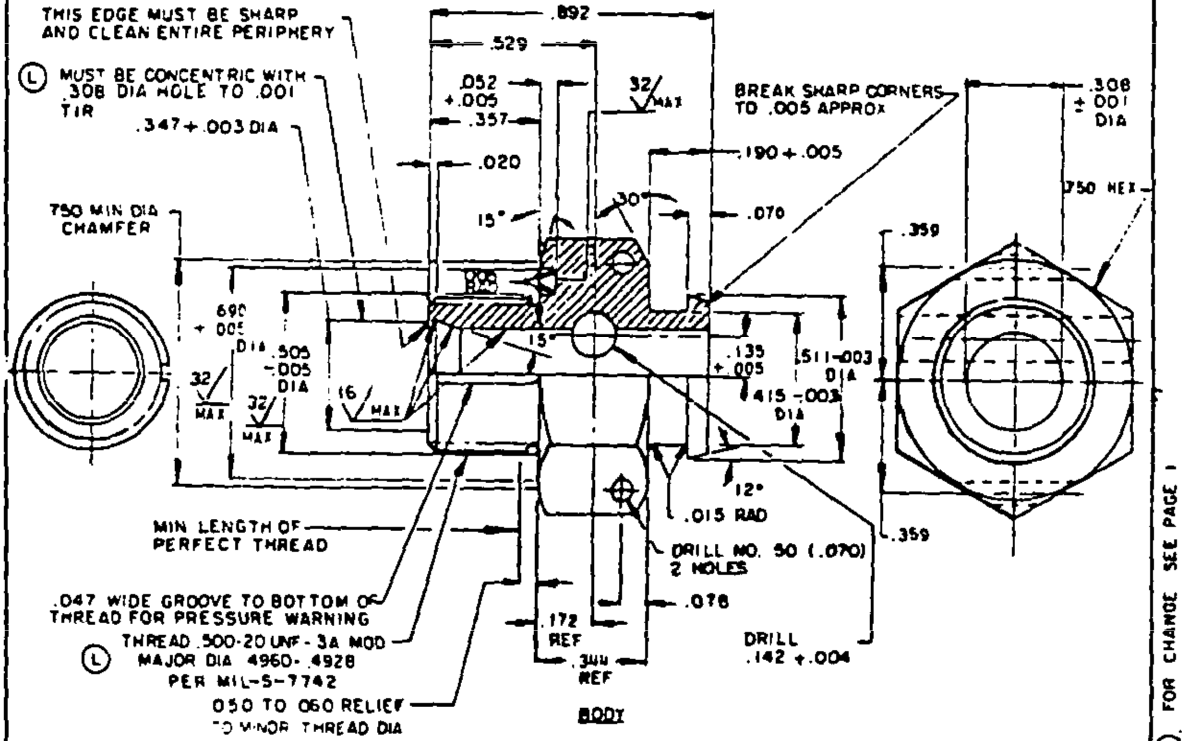
DD FORM 672-1

PROJECT NUMBER
1650-0419

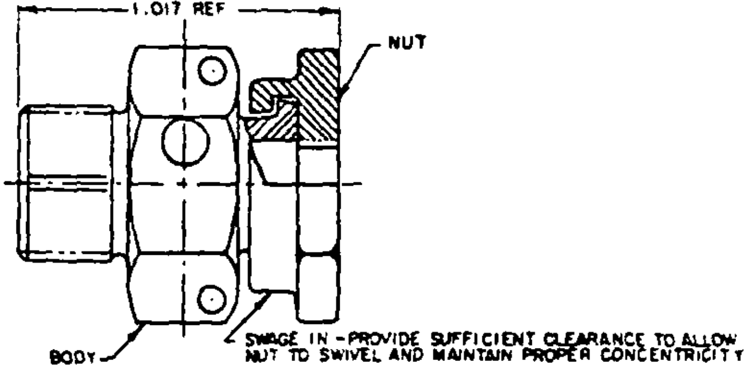
FED. SUP CLASS
4820


.020 RAD
.125 + .010
.100
.045 -.005
26.5°
.515 + .003 DIA
.010 RAD
THREAD .311-32
PITCH DIA .290 + .002
.090 +.005
.165
BREAK SHARP CORNERS .016 RAD
.750 HEX
MANUFACTURER'S NAME OR TRADEMARK AND MS28889-2 MUST BE PERMANENTLY STAMPED ON THE NUT

NUT


THIS EDGE MUST BE SHARP AND CLEAN ENTIRE PERIPHERY
Ⓛ MUST BE CONCENTRIC WITH .308 DIA HOLE TO .001 TIR
.347+.003 DIA
.892
.529
.052 +.005
.357
32/MAX
BREAK SHARP CORNERS TO .005 APPROX
.308 ± .001 DIA
.020
.190 + .005
15°
30°
.070
750 MIN DIA CHAMFER
.359
750 HEX
.690 + .005 DIA
.505 -.005 DIA
32/MAX
32/MAX
16/MAX
15°
.135 +.005
.511 -.003 DIA
.415 -.003 DIA
12°
.015 RAD
.359
MIN LENGTH OF PERFECT THREAD
DRILL NO. 50 (.070) 2 HOLES
.047 WIDE GROOVE TO BOTTOM OF THREAD FOR PRESSURE WARNING
.078
Ⓛ THREAD .500-20 UNF-3A MOD MAJOR DIA .4960-.4928 PER MIL-S-7742
.172 REF
.344 REF
DRILL .142 +.004
.050 TO .060 RELIEF TO MINOR THREAD DIA

BODY


1.017 REF
NUT
BODY
SWAGE IN - PROVIDE SUFFICIENT CLEARANCE TO ALLOW NUT TO SWIVEL AND MAINTAIN PROPER CONCENTRICITY BETWEEN NUT AND STEM WHEN ASSEMBLED

BODY AND NUT

REVISED

Ⓛ FOR CHANGE SEE PAGE 1

APPROVED

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

AMSC N/A

USER ACTIVITIES

REVIEWER ACTIVITIES

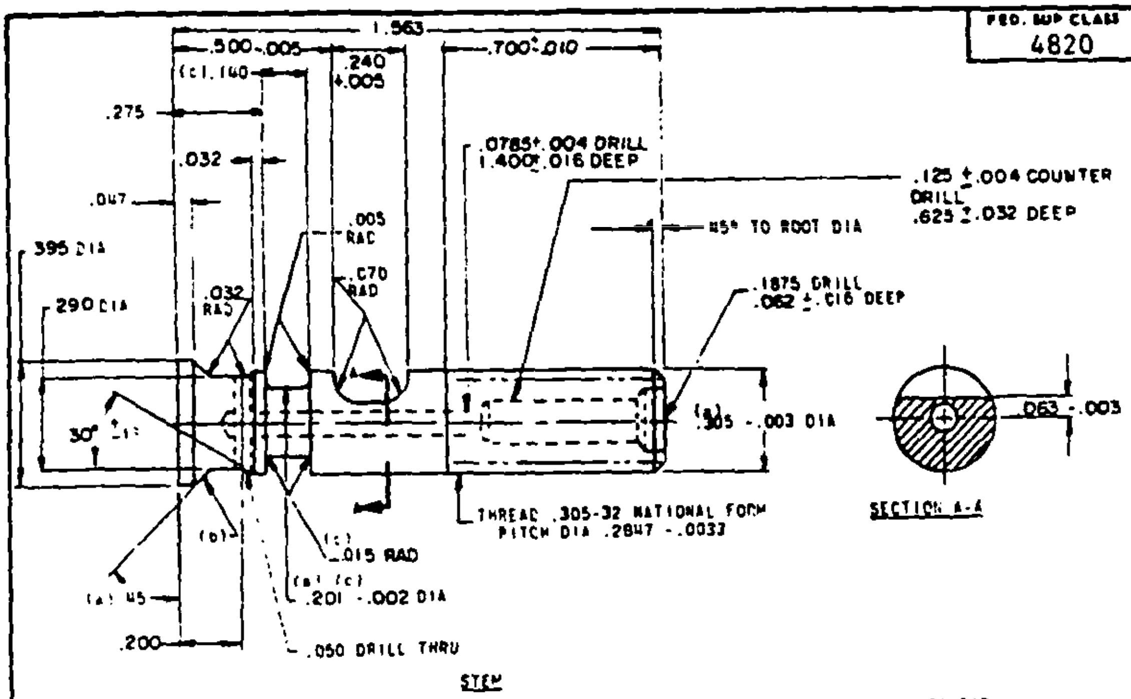
This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

P.A. USAF-71 Other Cust ARMY AV NAVY AS USAF-99	TITLE VALVE, AIR, HIGH PRESSURE CHARGING, 5000 PSI	MILITARY STANDARD MS28889
PROCUREMENT SPECIFICATION MIL-V-E-C-	SUPERSEDES	SHEET 2 OF 3

DD FORM 672-1

1650-0419

FED. SUP CLASS
4820

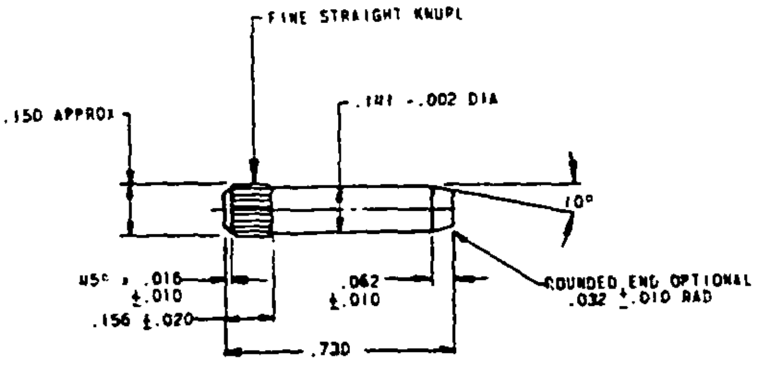

1.563
.500-.005
.700±.010
(c) 1.140
.240
±.005
.275
.032
.047
.005
RAD
.0785±.004 DRILL
1.400±.016 DEEP
.125±.004 COUNTER DRILL
.625±.032 DEEP
45° TO ROOT DIA
.395 DIA
.290 DIA
.070
RAD
.032
RAD
.1875 DRILL
.062±.016 DEEP
30° ±1°
(a)
.305 -.003 DIA
.063 -.003
SECTION A-A
THREAD .305-32 NATIONAL FORM
PITCH DIA .2847 -.0033
(b)
(c)
.015 RAD
(a) (c)
.201 -.002 DIA
(a) 45
.200
.050 DRILL THRU

STEM

(a) SURFACES INDICATED BY THESE DIMENSIONS SHALL BE CONCENTRIC TO EACH OTHER WITHIN .001 TIR

(c) THIS SURFACE SHALL BE FREE FROM BURRS, LONGITUDINAL AND SPIRAL TOOL MARKS AND SHALL BE SMOOTH EXCEPT THAT ANNULAR TOOL MARKS UP TO 63 RHR MAX WILL BE ALLOWED.

(b) SURFACE ROUGHNESS 32RHR MAX.


FINE STRAIGHT KNURL
.181 -.002 DIA
.150 APPROX
10°
45° x .016
±.010
.156 ±.020
.062
±.010
ROUNDED END OPTIONAL
.032 ±.010 RAD
.730

PIN RETAINER

USER ACTIVITIES

REVIEWER ACTIVITIES

REVISED (L) FOR CHANGE SEE PAGE 1

APPROVED

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited

AMSC - N/A

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selections for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

P.A. USAF-71 Other Cust ARMY-AT NAVY-AS USAF 99	TITLE VALVE, AIR, HIGH-PRESSURE CHARGING, 5000 PSI	MILITARY STANDARD MS28889
PROCUREMENT SPECIFICATION MIL-V-C164	SUPERSEDES	SHEET 3 OF 3

DD FORM 672-1 (Coordinated)

1650-0419